25900745

MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) REGULAR /IMPROVEMENT/ REAPPEARANCE EXAMINATION NOVEMBER 2025
FIRST SEMESTER

Discipline Specific Core Course (DSC) - MG1DSCBIF100-FUNDAMENTAL IT FOR BIOINFORMATICS (2024 ADMISSION ONWARDS)
Scheme

Duration: 2 Hours **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(1) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

*Short Answer Questions, Answer any **5** Questions, Each Question carries **2** marks*

1.List two common network protocols. [K]/[CO2]
- -HTTP – transfers web pages. (1 mark)
- -FTP – used for file transfer. (1 mark)

2. Identify three common input devices. [K]/[C01]
- Any two from Keyboard, Mouse,Scanner ,Microphone, Joystick (acceptable alternatives) with one line description (1 mark for each).

3. Differentiate between hardware and software. [U]/[C01]
- -Hardware: Physical components of a computer, examples (1 mark)
- -Software: Programs/instructions that operate hardware, examples (1 mark)

4. Define XML. [K]/[CO3]
- -XML = Extensible Markup Language. (1 mark)
- -Used to store, structure and transport data using custom tags. (1 mark)

5. Name the four main categories of operators in JavaScript. [K]/[CO3]
- Any four (0.5 marks each):
- Arithmetic
- Comparison
- Logical
- Assignment

6. Outline the syntax for a for loop in JavaScript. [K]/[CO3]
- Correct syntax (1.5 marks):

```
for (initialization; condition; increment) {
    // statements
    }
```

- Purpose : Used for repeated execution. (0.5 marks)

7.Outline basic application of bioinformatics in agriculture. [K]/[CO4]

- Any two (1 mark each):
- Identifying genes for high yield or disease resistance.
- Developing improved crop varieties.

[5x2 = 10]

Part B

*Short Essay Type Questions, Answer any **4** Questions, Each Question carries **5** marks*

8. Explain the various network topologies. [K]/[C02]

- Definition (1 mark)
- Features of any four topologies: Bus, Star, Ring, Mesh, Hybrid (1 mark each)

9. Explain the concept of multitasking and how it is implemented in different operating systems [U]/[CO1]

- -Definition (1 mark)
- -Types (3 marks):
- Pre-emptive multitasking (1.5 marks)
- Cooperative multitasking (1.5 marks)
- -Benefits/examples (1 mark)

10. Define an interpreter and explain its main purpose [U]/[C03]

- -Definition (2 marks): Executes code line by line.
- -Purpose (3 marks): Debugging, portability, used in Python, JS.

11. List some common CSS properties for text formatting. [U]/[CO3]

- -Any five : font-size, font-family, color, font-style, font-weight, text-align, text-decoration. (1 mark each)

12 List out basic terminologies in bioinformatics. [K]/[C04]

- -Any five (1 mark each):
- -Genome, BLAST, alignment, phylogeny, annotation, biological databases, protein structure.

13. Discuss major milestone in human genome project. [U]/[CO4]

- -Description (1 mark)
- -Any four milestones :Began 1990, completed 2003, Chromosomal mapping, Gene identification, Large databases, Advances in sequencing. (1 mark each)

[4x5 = 20]

Part C

*Long Essay Type Questions, Answer any **2** Questions, Each Question carries **10** marks*

14. Discuss the different types of IP addresses (IPv4, IPv6) and their implications. [K]/[CO2]

- -Definition (3 marks)
- -The different types of IP addresses.(5 marks)
- -Implications (2 marks)

15. Differentiate between high level and low level languages in detail. [U]/[CO3]

- -Definitions (2 marks)
- -Five comparisons (6 marks)

- -Examples (2 marks)

16. Show how to create internal and external links using the href attribute. [U]/[C03]
    - -Definition of hyperlink (2 marks)
    - -Internal link (3 marks)
    - -Internal page link (2 marks)
    - -External link (3 marks)
17. Explain why bioinformatics is known as an interdisciplinary field. [U]/[CO4]
    - -Definition (2 marks)
    - -Any six fields (6 marks)
    - -Conclusion (2 marks)

[2x10 = 20]